Hierarchical Convexity-Run Decomposition: Finite-Sample Chord-Lobe Recovery and Interpretable LC–MS Transfer

Saveliy Baturin
Independent Researcher

Draft August 26, 2026

Abstract

We introduce Hierarchical Convexity-Run Decomposition (HCRD), a finite nonlinear transform that replaces runs of one discrete-curvature sign by endpoint chords and recurses on retained knots. Its signed residuals form an exactly reconstructing, interpretable lobe hierarchy with linear total knot-only work. On an explicit alternating chord-lobe class under Gaussian noise, we prove first-level boundary and component recovery, including approximate sampled joins: if active curvature γ, join departure η, and the simultaneous noise threshold τ satisfy $\gamma > \eta + 2\tau$, every boundary is recovered with probability at least $1 - \delta$. Across 206,000 simulated signals, zero failures were observed among 80,000 draws in the strict certified regions. Structural results establish finite termination, affine equivariance, variation control, global discontinuity of the hard map, and stable companions. In frozen no-refit transfers between two fully labelled LC–MS studies, HCRD-8 plus an independently reimplemented quality score improved average precision by 0.1181 and 0.1005. An equal-dimensional Gaussian-derivative control tied HCRD in one direction, whereas HCRD was superior in the reverse; a larger TARDIS/FAME shift did not confirm superiority. HCRD therefore provides a simple morphological inductive bias with a quantitative recovery regime and explicit application boundaries.

1 Introduction

Empirical mode decomposition (EMD) motivated data-adaptive analysis of nonstationary signals through extrema and envelope sifting [10]. Intrinsic time-scale decomposition (ITD) subsequently introduced an efficient iterative baseline–rotation construction based on piecewise-linear transformations between extrema [8, 9]. In particular, its patent measures a lobe height from an extremum to the line joining neighbouring extrema and recursively decomposes the concatenated baseline. Thus straight-line lobe geometry and recursive piecewise-linear baselines are prior art. The method studied here arose independently from a different selection and update rule: partition a sampled graph into maximal runs of one divided-curvature sign, replace each entire run by its endpoint chord, and allow only the retained knots at the next level. Figure 1 previews the resulting curvature-run split, endpoint chords, and exact two-level synthesis.

The primitive is related to inflection-point sampling [11], but iteration produces a nested, exactly reconstructing hierarchy. More directly, the PDE sifting construction of Deléchelle et al. [6] converges to a piecewise-cubic interpolant through successive inflection points, while the α-shape construction of Chessel et al. [4] defines a convexity-driven geometric scale space. Recent EMD software also includes inflection-point interpolation as one of several local-mean estimators [21], and special-inflection-point EMD augments extrema with inflections from the signal and its derivatives [12]. These works preclude treating the use of inflection points itself as novel. HCRD differs in using a one-pass discrete curvature-sign walk, endpoint chords, and recursively nested eligible knot sets, rather than auxiliary envelope knots, PDE evolution, α-shape subsampling, or repeated IMF sifting.

Table 1: Closest one-dimensional representations. "Exact" means that the native components plus the final baseline reconstruct the input; costs are nominal.

Method	Primitive and native components	Reconstruction / hierarchy	Computation and regularity
HCRD	Curvature-sign runs; endpoint-chord residual lobes	Exact; nested knot subsets	$O(n)$ total sparse; hard decisions; irregular x
ITD [9]	Extrema updates; linear baseline and rotations	Exact; recursive baseline	$O(n)$ per level; hard extrema; ordered grid
LULU/DPT [1, 15]	Order / flat zones; constant pulses	Exact; size hierarchy	$O(n)$ Roadmaker; order-based sequence
RDP [7]	Maximum chord error; polygonal approximation	Exact with residual; split tree	$O(n^2)$ worst case; hard ties; arbitrary coordinates
ℓ_1 trend filtering [13]	Penalized curvature; piecewise-linear fit	Exact with residual; no native hierarchy	Convex optimization; continuous fit; irregular variants
Gaussian scale space [2]	Convolution scale; smoothed signals	Scale differences; scale-ordered	FFT / direct; continuous; regular-grid standard

The closest theoretical comparison classes further include Gaussian scale space [2], ℓ_1 trend filtering [13], iterative filtering [16], connected operators represented by region trees [19], and the LULU discrete pulse transform and its linear-time one-dimensional Roadmaker construction [1, 15]. Connected operators prune flat-zone region trees, while DPT produces constant block pulses on connected supports; neither uses affine residual lobes from the HCRD curvature-run knot tree. HCRD's contribution is the particular finite discrete operator and the combination of hierarchy, reconstruction, geometric sign, stability, limitation, and calibrated-inference results—not the generic idea of using curvature, inflection points, chord lobes, or a nonlinear component hierarchy.

Table 1 makes the closest distinctions explicit. ITD is the nearest chord-baseline recursion and DPT the nearest exact nonlinear pulse hierarchy; neither uses curvature-run knot selection. Trend filtering instead solves a stable penalized optimization problem.

The LC–MS use case is also distinct from end-to-end peak picking. PeakBot detects local maxima and classifies two-dimensional chromatographic regions with a convolutional network [3]; PeakDetective combines an autoencoder with active learning for dataset-specific curation [20]; TMSQE learns targeted peak-quality embeddings [23]; and MassCube integrates feature detection, evaluation, grouping, and annotation [24]. HCRD instead supplies transparent features for already bounded one-dimensional EICs and is tested here under no-target-refit transfer. These systems therefore define the application landscape but are not interchangeable baselines for the present fixed-box question.

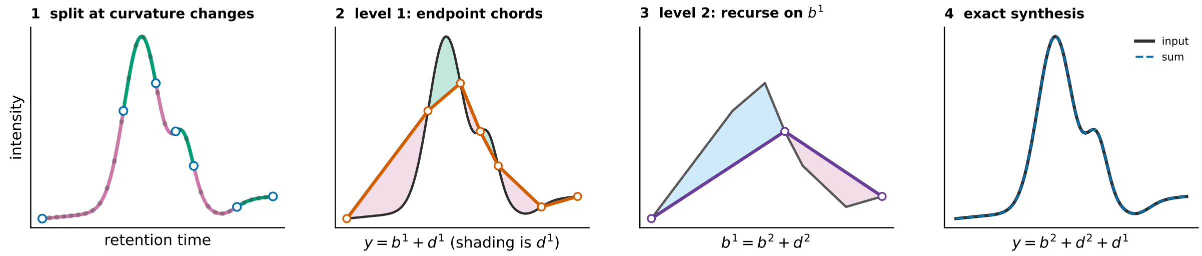

Figure 1: Two explicit HCRD steps on a synthetic EIC-like graph. Curvature-sign changes delimit retained knots. Replacing every level-1 run by its endpoint chord gives baseline b^1 and signed residual d^1; applying the same operation to b^1 gives b^2 and d^2. The displayed truncation reconstructs $y = b^2 + d^2 + d^1$ exactly, and recursion may continue. Shaded lobes can be summarised by area or energy, but the decomposition is the complete residual shape. This computed example illustrates the operator; it is not a stability proof and levels are not asserted to be frequency bands.

Our contributions are:

1. a complete finite algorithm for the centred curvature-run operator, with nested-hierarchy, exact-reconstruction, sparse linear-work, affine, variation, and interpretable lobe-shape results;
2. an HCRD-specific finite-sample recovery theorem for an explicit alternating chord-lobe class with exact or approximate sampled joins, local and stochastic hierarchy-agreement guarantees, and theorem-linked phase experiments over 206,000 signals;
3. a sharp global discontinuity boundary together with stable proximal and persistence companions;
4. frozen cross-study LC–MS validation, multilevel and area ablations, a matched-capacity conventional control, operational review-queue analysis, implementation and refitting sensitivities, and external limitation studies. Optional scan and post-selection theory is given in the supplement.

2 Definition of the transform

Let $x_0 < \cdots < x_{n-1}$ and $y = (y_0, \ldots, y_{n-1}) \in \mathbb{R}^n$. Define divided slopes and discrete curvatures by

$$s_i(y) = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}, \qquad \kappa_i(y) = s_i(y) - s_{i-1}(y). \tag{1}$$

For absolute and relative tolerances $a, r \geq 0$, define

$$\tau = a + r \max\{1, \max_i |s_i(y)|\}. \tag{2}$$

A curvature is active when $|\kappa_i| > \tau$ and inactive otherwise. The mathematical hard operator uses $a = r = 0$; the implementation defaults to a tiny relative tolerance to suppress floating-point curvature on interpolated affine segments. Algorithm 1 specifies every branch of the centred rule.

Algorithm 1 One centred HCRD hierarchy on an irregular grid

Input: strictly increasing x, samples y, tolerances $a, r \geq 0$, and optional depth cap L. Initially $E = (0, \dots, n-1)$.

1. On the current eligible list $E = (e_0, \dots, e_{m-1})$, set $z_i = y_{e_i}$, $u_i = x_{e_i}$, compute divided curvatures, and compute τ from the current divided slopes. If $m = 2$, stop.
2. Set the local start $p = 0$ and the new local knot list $K = (0)$.
3. Let f be the first active curvature index in $\{p+1, \dots, m-2\}$. If none exists, append $m-1$ to K and finish this level. Otherwise set the active sign to $\operatorname{sign}(\kappa_f)$, $\ell = f$, and scan right.
4. Ignore inactive curvatures. On each active curvature of the same sign, update ℓ. Let q be the first active curvature of the opposite sign. If no such q exists, append $m-1$ and finish the level.
5. If $q - \ell = 2$, the single inactive curvature at $\ell + 1$ is the sampled transition and the candidate is $c = \ell + 1$. At zero tolerance this is exactly the sampled-zero case.
6. Otherwise set $c = \ell$ only when $|\kappa_\ell| < |\kappa_q|$ and $\ell \neq f$; set $c = q$ in every other case. Thus equal magnitudes are tied to the right, first-opposite side, and a long inactive plateau never creates an interior knot.
7. Enforce strict coarsening by replacing c with $\max\{p+2, \min(c, m-1)\}$. Append c to K, set $p = c$, and return to step 3. The final remainder always closes at $m-1$.
8. Map the local knots back to original indices, $E' = (e_k : k \in K)$. Interpolate the endpoint chords through (x_i, y_i) for $i \in E'$ to obtain the next baseline; subtract it to obtain the signed detail.
9. Set $E \leftarrow E'$ and repeat until $|E| = 2$ or the optional cap L is reached. Later levels therefore use only previously retained knots.

Let $K(y)$ denote the first mapped knot set and let Ty be linear interpolation through (x_k, y_k) for $k \in K(y)$.

Set $b^0 = y$ and recursively define

$$b^{j+1} = Tb^j, \qquad d^{j+1} = b^j - b^{j+1}. \tag{3}$$

The implementation restricts eligible knots at level $j+1$ to knots retained at level j. This prevents the insertion of geometrically redundant samples inside an existing affine segment. In local eligible-knot coordinates, a nonterminal centred interval is required to advance by at least two old intervals; the first-opposite and sampled-zero cases below satisfy this convention explicitly.

3 Structural results

Theorem 1 (Exact reconstruction)**.** *For every computed level J,*

$$y = \sum_{j=1}^{J} d^j + b^J. \tag{4}$$

Proof. Substitute $d^j = b^{j-1} - b^j$ and telescope. □

Theorem 2 (Finite nested hierarchy)**.** *The eligible knot sets are nested. Every nonterminal HCRD interval consumes at least two intervals of the preceding level. Consequently the hierarchy ends at the endpoint chord after at most* $\max\{1, \lceil \log_2(n-1) \rceil\}$ *implemented levels.*

Proof. Eligibility gives nesting. A run can close only after the walk has passed the first active old curvature and reached either an eligible intervening zero or the first active curvature of opposite sign. Thus an old interior knot is removed in each nonterminal new interval. Pairing old intervals gives the halving recurrence, with at most one terminal remainder. □

Corollary 1 (Linear-work knot-only hierarchy). *Let $N_0 = n - 1$ and let N_j be the number of intervals after level j. An implementation that scans only the current eligible knot list has $O(n)$ total knot-walk work and stores*

$$\sum_{j=1}^{J} |K_j| < 2(n-1) + J = O(n). \tag{5}$$

Dense length-n baselines and details can be reconstructed exactly on demand.

Proof. The halving recurrence gives $N_j \le \lceil N_0/2^j \rceil$ and $J = O(\log n)$. For integer $N_0 \ge 1$, $\lceil N_0/2^j \rceil \le (N_0 - 1)/2^j + 1$; hence $\sum_{j=1}^{J} N_j < N_0 - 1 + J \le 2N_0$. Since $|K_j| = N_j + 1$, the displayed finite bound follows. The analogous series beginning with N_0 bounds the total scanned eligible intervals by $O(n)$. Chord interpolation through each stored knot set reproduces the dense baseline, and adjacent differences reproduce the details. □

Proposition 1 (Signed structures). *On a discretely convex run, $y - Ty \le 0$. On a discretely concave run, $y - Ty \ge 0$.*

Proof. A convex polygonal graph lies below its endpoint secant. Reverse the inequality for a concave graph. □

For structure r of detail d^j on $I_{jr} = [x_a, x_b]$, define its signed area, geometric mass, and quadratic residual energy by

$$S_{jr} = \int_{I_{jr}} d^j(x)\,dx, \qquad A_{jr} = \int_{I_{jr}} |d^j(x)|\,dx, \qquad E_{jr} = \int_{I_{jr}} d^j(x)^2\,dx, \tag{6}$$

where d^j is the exact piecewise-linear interpolant of the sampled residual. These quantities are computable from retained knots without materializing a length-n detail. Signed structures give $A_{jr} = |S_{jr}|$. Under $y \mapsto cy + \alpha + \beta x$, A_{jr} scales by $|c|$ and E_{jr} by c^2; under an orientation-preserving horizontal affine change $x \mapsto \rho x + \tau$, both integrals acquire the Jacobian factor ρ. Thus A_{jr} is a literal morphology measure in signal-units times coordinate-units. It is not spectral energy, and the nonorthogonal hierarchy has no Parseval identity.

Proposition 2 (Dimensionless lobe shape). *For a nonzero structure let $W = x_b - x_a$, $H = \max_{I_{jr}} |d^j|$, and define*

$$F = \frac{A_{jr}}{WH}, \qquad C = \frac{A_{jr}^2}{WE_{jr}}. \tag{7}$$

Then $0 < F \le 1$ and $0 < C \le 1$. Both are invariant under nonzero vertical scaling, affine trend addition, and orientation-preserving horizontal affine changes. A piecewise-linear triangular lobe has $(F, C) = (1/2, 3/4)$ regardless of apex location; a parabolic lobe $4Ht(W - t)/W^2$ has $(F, C) = (2/3, 5/6)$.

Proof. $A \le WH$ follows from $|d| \le H$, while Cauchy–Schwarz gives $A^2 \le WE$. The scaling laws above cancel in both ratios. Direct integration gives the two examples. □

The first ratio is the implemented per-structure shape factor. The second is available as a quadratic concentration descriptor but was not added after the LC–MS protocols were frozen. Keeping A_{jr} or either invariant for each moving support and level also yields a low-dimensional temporal morphology series; collapsing all structures to one total mass discards that evolution.

Theorem 3 (Affine equivariance). *For zero curvature tolerance, nonzero a, and $\ell_i = \alpha + \beta x_i$,*

$$T(ay + \ell) = aTy + \ell. \tag{8}$$

Hence every HCRD detail is invariant to affine trend addition and scales by a.

Proof. Affine addition cancels from adjacent divided-slope differences. Multiplication by a preserves all transition locations, although negative a reverses the signs. Chord interpolation commutes with both operations. $\square$

Theorem 4 (Range and total variation)**.** *For one HCRD step,*

$$\min y \leq (Ty)_i \leq \max y, \qquad TV(Ty) \leq TV(y). \tag{9}$$

Proof. Every chord sample is a convex combination of its endpoint values. Its variation equals the absolute difference between the endpoint values, which is no greater than the variation of the removed polygonal subpath. $\square$

4 Stability and its obstruction

Theorem 5 (Stability inside a curvature-decision cell)**.** *On a unit grid suppose $\gamma = \min_i |\Delta^2 y_i| > 0$ and let*

$$\eta = \min_{\Delta^2 y_i \Delta^2 y_{i+1} < 0} \big||\Delta^2 y_i| - |\Delta^2 y_{i+1}|\big|, \tag{10}$$

with $\eta = +\infty$ if there is no sign transition. For the centred rule, if $\|e\|_\infty < \min\{\gamma/4, \eta/8\}$, then $K(y+e) = K(y)$ and

$$\|T(y+e) - Ty\|_\infty \leq \|e\|_\infty, \tag{11}$$

$$\|(I-T)(y+e) - (I-T)y\|_\infty \leq 2\|e\|_\infty. \tag{12}$$

Proof. The coefficient norm of the second-difference stencil $(1, -2, 1)$ is four, so the first margin prevents a curvature-sign change. A difference of two absolute curvature magnitudes moves by at most $8\|e\|_\infty$, so the second margin preserves every centred transition comparison. The knot walk is therefore fixed. Fixed-knot interpolation is a convex combination of endpoint perturbations, and the detail estimate follows by the triangle inequality. $\square$

The second margin is necessary: $(0,0,1,3,4)$ has curvatures $(1,1,-1)$, whereas $(0,0,1,3-\varepsilon, 4-2\varepsilon)$ has the same signs but selects the other side of the tied centred transition. The legacy first-opposite rule does not compare magnitudes and requires only the sign margin.

Proposition 3 (Global discontinuity)**.** *The raw HCRD baseline operator is not globally continuous.*

Proof. For $y^\pm = (-2, 0, 2 \pm \varepsilon, -2)$, the inputs approach one another as $\varepsilon \downarrow 0$, whereas one first baseline retains the interior peak and the other becomes the constant endpoint chord. Their sup-norm separation tends to four. $\square$

Theorem 6 (No continuous generic-agreement extension)**.** *For four-sample signals let $G = \{y : \min_i |\Delta^2 y_i| > 0\}$. There is no globally continuous map $F : \mathbb{R}^4 \to \mathbb{R}^4$ that agrees with the raw HCRD baseline T on all of G.*

Proof. For every $0 < \varepsilon < 1$, both $y^\pm_\varepsilon = (-2, 0, 2 \pm \varepsilon, -2)$ lie in G and converge to the same y^0. Agreement on G makes $F(y^\pm_\varepsilon)$ equal the two HCRD outputs in the preceding counterexample, whose limits are separated by four. Continuity at y^0 would force both limits to equal $F(y^0)$, a contradiction. $\square$

Remark 1. *HCRD is not an orthogonal transform and need not contract Euclidean energy. For $y = (-2, -1, -2)$, one has $Ty = (-2, -2, -2)$ and $\|Ty\|_2 > \|y\|_2$.*

5 A stable proximal companion

Let D be the divided-curvature matrix and let ϕ be a proper closed convex function. Define

$$P_\phi y = \operatorname*{argmin}_z \left\{ \frac{1}{2}\|y - z\|_2^2 + \phi(Dz) \right\}, \qquad Q_\phi = I - P_\phi. \tag{13}$$

We implement the established ℓ_1 trend penalty $\phi(u) = \lambda\|u\|_1$ [13] and a quadratic curvature penalty $\phi(u) = \lambda\|u\|_2^2/2$. These guide operators themselves are not claimed as novel; their role is to give HCRD an exact stable outer split.

Theorem 7 (Exact globally nonexpansive outer split)**.** *Both P_ϕ and Q_ϕ are firmly nonexpansive and*

$$y = P_\phi y + Q_\phi y. \tag{14}$$

Moreover, if $D\ell = 0$, then $P_\phi(y + \ell) = P_\phi y + \ell$ and $Q_\phi(y + \ell) = Q_\phi y$.

Proof. The divided-curvature matrix has full row rank, so properness of ϕ makes $\phi \circ D$ proper. The objective is coercive and strongly convex, hence has a unique minimizer. The standard proximal-map theorem and Moreau decomposition make a proximal map and its residual firmly nonexpansive [18]. Exact splitting is the definition of Q_ϕ. Translating both y and the candidate minimizer by $\ell \in \ker D$ leaves the objective unchanged. □

Corollary 2 (Irregular-grid sign certificate)**.** *Suppose $\|e\|_2 \le \rho$ and put $h_{i-1} = x_i - x_{i-1}$ and $h_i = x_{i+1} - x_i$. The ith guide curvature changes by at most*

$$c_i\rho = 2\rho(h_{i-1}^{-1} + h_i^{-1}). \tag{15}$$

Thus every guide curvature satisfying $|\kappa_i(P_\phi y)| > c_i\rho$ has a fixed sign throughout the input ball. Other curvatures must be marked uncertified.

Proof. Theorem 7 bounds the guide perturbation in ℓ_2, hence also coordinatewise, by ρ. The absolute coefficient sum of the irregular three-point curvature row is $2(h_{i-1}^{-1} + h_i^{-1})$. □

We retain $Q_\phi y$ explicitly and apply HCRD only to the guide $P_\phi y$, so the complete representation still reconstructs y. Theorem 7 applies to the outer split, not to the hard knot hierarchy. Indeed, Theorem 6 rules out global continuity for any hard replacement that preserves every generic raw decision.

5.1 Stable signed-curvature persistence

Hard knot locations are not the only possible summary of the curvature walk. Let P be the path on the $n-2$ interior sample indices and define the nonnegative vertex functions

$$u_y^+ = (D_x y)_+, \qquad u_y^- = (-D_x y)_+. \tag{16}$$

Extend them linearly over the path edges. For a nonnegative function u, let $\mathcal{D}_0^{\mathrm{fin}}(u)$ be the finite part of its ordinary zero-dimensional superlevel persistence diagram and let $M(u) = \max u$ record the birth of its unique essential component. Define

$$\begin{aligned} d_{\mathrm{SC}}(y,z) = \max\{ & d_B(\mathcal{D}_0^{\mathrm{fin}}(u_y^+), \mathcal{D}_0^{\mathrm{fin}}(u_z^+)), |M(u_y^+) - M(u_z^+)|, \\ & d_B(\mathcal{D}_0^{\mathrm{fin}}(u_y^-), \mathcal{D}_0^{\mathrm{fin}}(u_z^-)), |M(u_y^-) - M(u_z^-)|\}. \end{aligned} \tag{17}$$

This is the pullback of a metric on the signed persistence summaries and is therefore a pseudometric on signals. It becomes a metric on the quotient $y \sim z$ if and only if their two finite diagrams and two essential birth heights agree. Quotienting only by affine functions is insufficient: on a unit grid, $y = (0,0,1,2,5)$ and $z = (0,0,2,4,7)$ have positive curvatures $(1,0,2)$ and $(2,0,1)$, hence identical summaries by path reflection and $d_{\mathrm{SC}}(y,z) = 0$, although $z - y$ is not affine.

Theorem 8 (Global signed-curvature persistence stability)**.** *For two signals on the same irregular grid,*

$$d_{\mathrm{SC}}(y,z) \le C_x \|y-z\|_\infty, \qquad C_x = 2\max_i\{h_{i-1}^{-1} + h_i^{-1}\}. \tag{18}$$

If $\|y-z\|_\infty \le \varepsilon$, every finite bar of either sign with lifetime greater than $2C_x\varepsilon$ has a finite same-sign match for the other signal.

Proof. The absolute row sum of D_x is $2(h_{i-1}^{-1} + h_i^{-1})$, while positive and negative part maps are 1-Lipschitz. Hence $\|u_y^\pm - u_z^\pm\|_\infty \le C_x\|y-z\|_\infty$. The finite path is triangulable and the piecewise-linear functions are tame, so classical persistence-diagram stability applies [5]. The unique essential classes must match each other; the remaining matching is between finite points. A finite bar of lifetime L costs $L/2$ to match to the diagonal, proving the last statement. □

Peak indices are retained by the implementation as interpretive metadata but are not included in d_{SC}. Two distant curvature peaks of equal height can exchange the essential label under an arbitrarily small perturbation, so a global coordinate-stability statement would be false. Thus Theorem 8 is a stable lobe representation, not a concealed claim that the hard HCRD knots are continuous.

Proposition 4 (Curvature visibility limit)**.** *Let $y(t) = g(t) + a\sin(\omega t)$ on an interval and suppose $g''(t) \ge c > |a|\omega^2$ throughout that interval. Then $y''(t) > 0$ there, so the sinusoidal component creates no curvature-sign transition and cannot be isolated by a raw inflection-run rule on that interval.*

Proof. $y''(t) = g''(t) - a\omega^2\sin(\omega t) \ge c - |a|\omega^2 > 0$. Thus y is strictly convex; its divided slopes on every increasing sampled grid are increasing, so its discrete curvatures have one positive sign. □

6 A simultaneous noise-aware modification

Assume a uniform grid with spacing h and observations $Y_i = y_i + \varepsilon_i$, where the errors are independent $N(0,\sigma^2)$ variables.

Theorem 9 (Family-wise curvature threshold)**.** *For $0 < \delta < 1$, let*

$$\tau = \frac{\sigma}{h}\sqrt{12\log\frac{2(n-2)}{\delta}}. \tag{19}$$

With probability at least $1-\delta$, every observed curvature differs from its population value by at most τ. On this event, population zero curvatures are thresholded to zero and all curvatures with magnitude greater than 2τ retain their correct active sign.

Proof. Each curvature error has variance $6\sigma^2/h^2$. Apply a Gaussian tail bound to each of the $n-2$ entries and take a union bound; independence of curvature errors is not required. The classification statement follows from the triangle inequality. □

For a common physical-unit alternative to the row-normalized anomaly score, let $h_{\min} = \min_i(x_{i+1} - x_i)$,

$$\epsilon = \sigma\sqrt{2\log(2n/\delta)}, \qquad \eta = 4\epsilon/h_{\min}, \qquad \tau = \lambda\eta, \quad \lambda \ge 1, \tag{20}$$

run HCRD with absolute curvature tolerance τ, and set

$$Z(t) = \left[\max_j |d_j(t)| - 2\epsilon\right]_+. \tag{21}$$

Theorem 10 (Common-scale affine-null certificate)**.** *If $Y_i = \alpha + \beta x_i + e_i$ with independent $e_i \sim N(0, \sigma^2)$, then $\Pr\{Z(t) = 0$ for every $t\} \geq 1 - \delta$.*

Proof. A union bound gives $\|e\|_\infty \leq \epsilon$ with probability at least $1 - \delta$. On every retained grid a divided slope changes by at most $2\epsilon/h_{\min}$, so a curvature changes by at most $\eta \leq \tau$. The affine mean has zero curvature; the thresholded hierarchy therefore keeps only the endpoints. Both a noisy sample and its noisy endpoint chord lie within ϵ of their affine counterparts, hence every density is at most 2ϵ. □

Theorem 11 (Multilevel hierarchy-agreement margins)**.** *Run the noiseless minimum-curvature hierarchy with tolerance τ through level L. At every visited level suppose each curvature declared inactive satisfies $|\kappa_i| + \eta \leq \tau$, each active curvature satisfies $|\kappa_i| - \eta > \tau$, and each comparison $|\kappa_i| < |\kappa_j|$ used to choose a side of an unsampled sign transition has gap $||\kappa_i| - |\kappa_j|| > 2\eta$. Then, with probability at least $1 - \delta$, the noisy and noiseless knot sets agree through level L. Consequently a retained noiseless chord lobe of height $H > 4\epsilon$ has strictly positive score at its noiseless peak on that event.*

Proof. On $\|e\|_\infty \leq \epsilon$, every active-grid curvature moves by at most η. The first two margins preserve threshold status and sign; the final margin preserves every magnitude comparison. Thus the complete first-level branch trace is unchanged. Induction applies because equal knot sets expose the same retained grid at the next level. Under agreement, a density changes by at most 2ϵ; subtracting the null radius gives the peak lower bound $[H - 4\epsilon]_+$. □

Theorem 12 (Stochastic hierarchy agreement)**.** *Fix a grid, tolerance τ, and depth L. For a latent signal f, replay all threshold and centred magnitude-comparison decisions visited by its noiseless hierarchy. Let $M_{\text{th}}(f)$ be the smallest visited distance between a curvature magnitude and τ, let $M_{\text{cmp}}(f)$ be the smallest visited gap between compared curvature magnitudes, and define the sufficient input decision radius*

$$R(f) = \frac{h_{\min}}{4} \min\{M_{\text{th}}(f), M_{\text{cmp}}(f)/2\}, \tag{22}$$

with an empty minimum interpreted as $+\infty$. If random F satisfies $\Pr\{R(F) \leq r\} \leq H(r)$ and $e \sim N(0, \sigma^2 I_n)$ is independent of F, then

$$\Pr\{K_\ell(F + e) = K_\ell(F),\ 1 \leq \ell \leq L\} \geq \Big[1 - H(r) - 2n \exp\{-r^2/(2\sigma^2)\}\Big]_+. \tag{23}$$

The bound holds for every $r > 0$ and may therefore be optimized over r.

Proof. A sup-norm perturbation u changes every visited divided curvature by at most $4u/h_{\min}$ and a compared magnitude gap by at most twice that quantity. Thus $\|e\|_\infty < R(F)$ preserves the complete visited branch trace, and induction preserves every knot set. Agreement holds on $\{R(F) > r\} \cap \{\|e\|_\infty < r\}$; a union bound and the Gaussian maximum tail give the result. □

Corollary 3 (Transverse curved-background–lobe populations)**.** *Suppose $F = F_\Theta$, where Θ has density at most $p_{\max}$ on a compact parameter box. Partition the box into finitely many hierarchy branch cells and write the signed decision margins, already scaled to input-radius units in the definition of R, as $m_j(\theta)$. If each m_j is C^1 on its cells, $\|\nabla m_j\|_2 \geq g_j > 0$ in $|m_j| \leq r_0$, and every level set $m_j^{-1}(u)$ has $(d-1)$-measure at most A_j for $|u| \leq r_0$, then*

$$H(r) = 2p_{\max} r \sum_j A_j/g_j, \qquad 0 < r \leq r_0, \tag{24}$$

is valid in Theorem 12.

Proof. The coarea formula bounds the probability of the r-tube around decision surface j by $2p_{\max} r A_j/g_j$; union-bound the finite visited margin list over the branch cells. □

The radius is sufficient, not largest. The corollary covers random lobe amplitude, location, width and asymmetry together with finitely parameterised curved backgrounds when decision surfaces are transverse. Atoms on a tie, tangencies, infinite-dimensional backgrounds, and same-data estimation of H remain outside the result.

7 Finite-sample recovery on a chord-lobe class

On a uniform grid $x_i = x_0 + ih$, define the divided-slope curvature

$$\kappa_i(z) = \frac{z_{i+1} - 2z_i + z_{i-1}}{h}, \qquad 1 \le i \le n-2. \tag{25}$$

This scaling matches Algorithm 1; it is a slope difference rather than a divided second derivative.

Theorem 13 (Finite-sample generative chord-lobe recovery)**.** *Let $0 = k_0 < k_1 < \cdots < k_J = n-1$, with $k_r - k_{r-1} \ge 2$, and write $f = b + q$. Suppose b is affine on each $[x_{k_{r-1}}, x_{k_r}]$, $q_{k_r} = 0$, and the population curvatures of f satisfy*

1. *$\kappa_{k_r}(f) = 0$ at every interior knot;*
2. *inside successive knot intervals they have alternating constant sign and magnitude at least $\gamma > 0$.*

Observe $Y_i = f_i + e_i$ with independent $e_i \sim N(0, \sigma^2)$ and fix $0 < \delta < 1$. Set

$$\tau_\delta = \frac{\sigma}{h}\sqrt{12 \log \frac{4(n-2)}{\delta}}, \qquad \epsilon_\delta = \sigma\sqrt{2 \log \frac{4n}{\delta}}. \tag{26}$$

Run the first centred HCRD step with absolute curvature tolerance τ_δ. If $\gamma > 2\tau_\delta$, then with probability at least $1 - \delta$ its knot set is exactly $\{k_0, \dots, k_J\}$. On the same event, for its chord baseline $\widehat{b}$ and detail $\widehat{q} = Y - \widehat{b}$,

$$\|\widehat{b} - b\|_\infty \le \epsilon_\delta, \quad n^{-1}\|\widehat{b} - b\|_2^2 \le \epsilon_\delta^2, \quad \|\widehat{q} - q\|_\infty \le 2\epsilon_\delta, \quad n^{-1}\|\widehat{q} - q\|_2^2 \le 4\epsilon_\delta^2. \tag{27}$$

Proof. With probability at least $1 - \delta/2$, every curvature error is at most τ_δ: this is Theorem 9 with its error budget replaced by $\delta/2$. Population zeros are then inactive, whereas $\gamma > 2\tau_\delta$ makes every active curvature remain active with the correct sign. Hence each declared knot is the unique inactive eligible point between opposite active blocks. The centred transition rule selects that point, and the two-segment gap prevents the strict-coarsening correction from moving it. Induction along the first-level walk proves exact knot recovery.

Independently, a Gaussian maximum bound gives $\|e\|_\infty \le \epsilon_\delta$ with probability at least $1 - \delta/2$. On exact recovery, $f_{k_r} = b_{k_r}$ and b is its own chord interpolant, so $\widehat{b} - b$ is the chord interpolant of the knot errors. Its sup norm is at most $\|e\|_\infty$; subtracting it from e gives the detail bound. The MSE bounds follow from the sup-norm bounds, and a union bound completes the proof. □

Corollary 4 (Recovery with approximate sampled joins)**.** *Retain the model, knot spacing, endpoint, active-block, and Gaussian-noise assumptions of Theorem 13, but replace its zero-join condition by the fixed structural bound*

$$|\kappa_{k_r}(f)| \le \eta, \qquad 1 \le r < J, \quad \eta \ge 0. \tag{28}$$

Run the first centred HCRD step with relative tolerance zero and any absolute tolerance t satisfying

$$\tau_\delta \le t \le \eta + \tau_\delta. \tag{29}$$

This interval includes both the original noise-only tolerance $t = \tau_\delta$, which does not require knowledge of η, and the join-inactivating tolerance $t = \eta + \tau_\delta$. If

$$\gamma > \eta + 2\tau_\delta, \tag{30}$$

then, with probability at least $1-\delta$, the knot set is exactly $\{k_0, \ldots, k_J\}$. The four baseline and detail bounds in Theorem 13 hold on the same event.

Proof. On the simultaneous curvature event from Theorem 13, each observed join has magnitude at most $\eta + \tau_\delta$, whereas every adjacent active curvature has magnitude at least $\gamma - \tau_\delta > \eta + \tau_\delta$, remains above t, and retains its population sign. If a join is inactive at t, the isolated-inactive rule selects it. If it is active, its sign agrees with either the left or right block. The sign transition then occurs immediately after or before the join; in both cases the centred minimum-magnitude rule selects the join because it is strictly smaller than the adjacent active curvature. The two-segment gap keeps the smaller left-side candidate past the first active curvature. Thus every declared knot is recovered. The sample-maximum event and interpolation argument are unchanged. □

The bound η is fixed independently of the observed noise when the corollary is used as a certificate. The noise-only implementation itself does not estimate or use η; a data-derived certificate would require a simultaneous upper confidence bound.

The assumptions are structural rather than circular: they describe sampled curvature before running HCRD. An explicit subclass has $k_r = rm$, a globally affine b, and alternating parabolic lobes with amplitudes $A_r > 0$,

$$q_i = s(-1)^r 4A_r u_i(1-u_i), \qquad u_i = (i - rm)/m, \quad rm \le i \le (r+1)m. \tag{31}$$

Here

$$\gamma = \frac{8 \min_r A_r}{m^2 h}, \qquad \eta = \frac{4(m-1)\max_r |A_r - A_{r-1}|}{m^2 h}. \tag{32}$$

Equal amplitudes give $\eta = 0$ and recover the theorem's exact sampled joins; variable amplitudes fall under Corollary 4 whenever its separation condition holds. When that separation fails, unequal adjacent amplitudes can move a boundary by one sample even without noise. The guarantees are sufficient, not necessary, separation conditions.

Optional inference extensions. Fixed or independently guided HCRD lobes can also be used as templates in finite or continuous scans, and independent replicates or buffered folds can separate structure selection from testing. Matching-order dictionary bounds, projected-norm calibration, replicate pivots, and nested-tree gatekeeping are developed in the supplement. They are optional inferential uses of HCRD rather than assumptions needed by the transform or recovery results below.

8 Experiments

The primary comparison protocol was frozen locally before inspecting results; it was not deposited in a public preregistration registry. It tests affine chord baselines with alternating convex/concave equal-amplitude parabolic lobes. Generic smoother hyperparameters are selected by an oracle against the known latent baseline separately for each trial. The primary endpoint is baseline mean squared error. A task-specific superiority statement requires a paired bootstrap interval below zero and a multiplicity-corrected exact sign test. Out-of-class tests include polynomial trends, crossing chirps, weak oscillations under strong background curvature, boundary outliers, irregular sampling, and non-Gaussian noise.

The theorem-linked R1 experiment varied $\rho = \gamma/\tau_\delta$ over 11 values on four fixed event-count/width configurations: $(J, m) = (2, 4), (4, 8), (8, 16), (16, 8)$. Each cell contained 1000 independent draws, for 44,000 trials in total. Figure 2 shows a clear transition. Exact recovery ranged from 0.433 to 0.879 across configurations at $\rho = 1.45$, from 0.980 to 0.991 at $\rho = 1.75$, and was 0.999 throughout at $\rho = 1.90$. In all eight cells above the strict theorem boundary $\rho > 2$, it was 1.000; the smallest cellwise 95% Wilson lower bound was 0.9962. The joint event of exact knots and both reconstruction bounds had minimum probability 0.9980 above the boundary (minimum Wilson lower bound 0.9927). The result validates a sufficient condition on its declared class, not a necessary boundary for arbitrary waveforms.

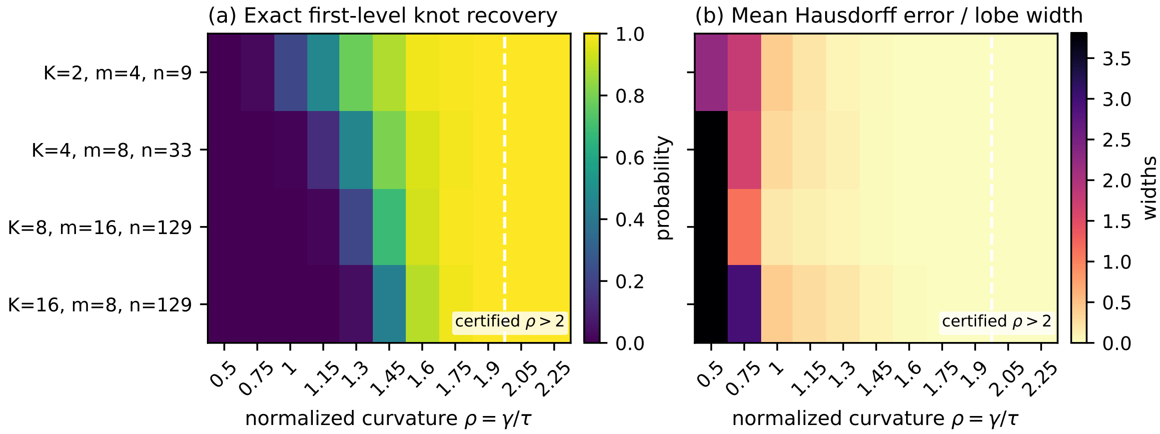

Figure 2: Theorem-linked recovery phase diagram under iid Gaussian noise. Rows vary lobe count J, samples per lobe m, and total length n; columns vary normalized active curvature $\rho = \gamma/\tau_\delta$. Left: exact first-level knot recovery. Right: symmetric Hausdorff boundary error divided by lobe width. The dashed line marks the strict sufficient region $\rho > 2$; all 8000 draws in that region recovered the exact knots.

Figure 3 maps departure from exact sampled joins. The corresponding experiment varied both η/τ_δ and γ/τ_δ over 162,000 independent signals with unequal adjacent lobe amplitudes. We compared the noise-only tolerance $t = \tau_\delta$ with the conservative join-inactivating choice $t = \eta + \tau_\delta$. All 72,000 draws in the strict certified region $\gamma/\tau_\delta > \eta/\tau_\delta + 2$ recovered every boundary for both choices (smallest cellwise 95% Wilson lower bound 0.9962), with no implication violation. Outside that region the noise-only choice was never worse cellwise and was often markedly better, confirming that η is useful for a certificate but need not be supplied to the algorithm.

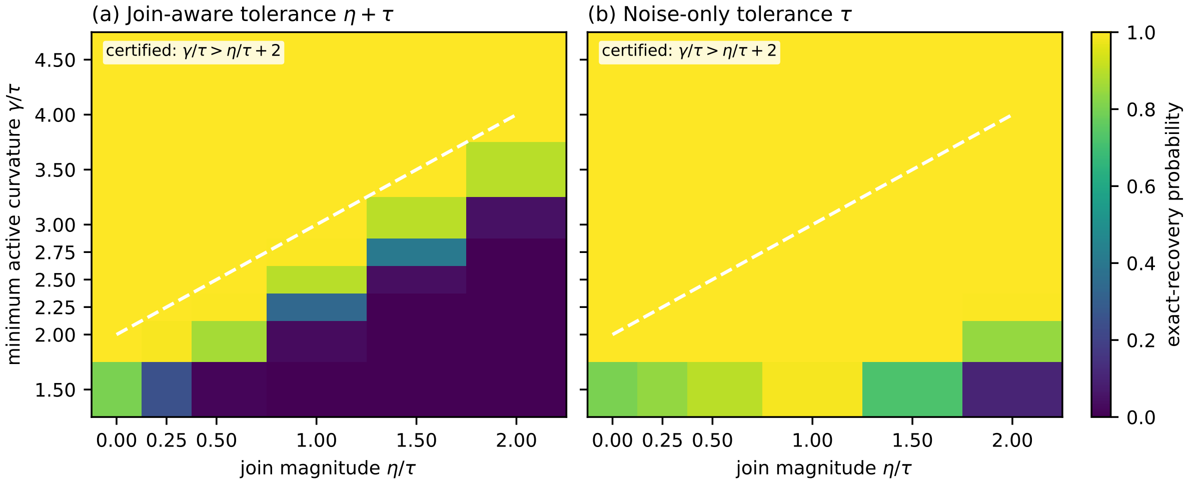

Figure 3: Departure from exact sampled joins. Exact first-level recovery is shown against active curvature and join departure, both normalized by the simultaneous noise threshold. The dashed line is the strict sufficient boundary $\gamma/\tau_\delta > \eta/\tau_\delta + 2$. Left: the conservative join-inactivating tolerance. Right: the original noise-only tolerance, which does not require knowledge of η.

8.1 Synthetic comparisons and computational audit

On the exact chord-lobe class, centred HCRD recovered the latent chord baseline exactly and outperformed oracle-tuned EMD, VMD, ITD, trend filtering, CEEMDAN, and Iterative Filtering baselines. This class-matched result illustrates the geometric distinction rather than global dominance over mode decompositions. With Gaussian noise, a plug-in MAD-thresholded variant retained lower mean baseline error than the same comparators in fresh-seed confirmations. Sparse knot-only storage preserved downstream features bitwise, reduced construction time 11.89-fold relative to dense materialisation, and independent-signal process parallelism reached 1.96-fold throughput speedup on 3840 signals. Complete settings, paired intervals, tables, stable-guide experiments, and the CWRU ceiling result are reported in the supplement.

8.2 Expert-labelled chromatographic lobes

An initial double-group holdout on the expert-labelled EIC corpus of Müller et al. [17] gave the largest AP to raw plus HCRD-8 (0.4092 versus 0.3820 for a frozen conventional morphology bank), but the primary two-way cluster interval crossed zero. Its ablations showed that the full signed multilevel decomposition, not polygon area alone, carried the useful signal. The complete E1 design and table are in the supplement. The independent E2 study below supplies the central cross-study evidence.

The independent E2 study then used the two fully expert-labelled HILIC studies of Kumler et al. [14]. Their two-parameter logistic quality score combines correlation with an idealized beta-shaped peak and a residual SNR. The source study found this simple model unusually robust under Falkor/MESOSCOPE transfer, making it a stronger and more interpretable primary comparator than a broad post-hoc model search. We used an independent global-window reimplementation of the published two-component score from the raw mzML feature boxes; on 1495 Falkor features its Pearson correlations with the authors' SNR and bell-correlation values were 0.906 and 0.789 (Spearman 0.871 and 0.805).

Before reading any raw waveform, we froze both transfer directions, Good versus Bad labels, exclusion of Ambiguous/standards-only cases, per-file robust aggregation, one identical standardized logistic learner, and 10,000 paired feature-bootstrap replicates. The classifier was fit on all 1821 unambiguous Falkor features and applied without refitting to 2193 MESOSCOPE features, then the direction was reversed. Figure 4 reports the result. HCRD-8 plus qscore improved AP over qscore from 0.7774 to 0.8955 in Falkor-to-MESOSCOPE and from 0.7984 to 0.8990 in the reverse direction. The paired differences were 0.1181 ($[0.0613, 0.1781]$) and 0.1005 ($[0.0559, 0.1470]$); both Holm-adjusted values were 0.00040. Every pre-specified success criterion was met.

This conclusion was insensitive to four fixed qscore implementations. We changed the minimum EIC length from eight to the authors' five-point rule and tested two-, five-, and seven-variable across-file summaries. HCRD-8 added 0.0765–0.1517 AP across the resulting eight direction–implementation contrasts, and every paired 95% interval remained above zero. The strongest qscore-only variants reached AP 0.8107 and 0.8221, while their corresponding HCRD combinations reached 0.8954 and 0.8986. On the 1277 unambiguous author-keyed Falkor features, the current median SNR and correlation summaries had Pearson correlations 0.934 and 0.867 with the author outputs. Full implementation-sensitivity results are in the supplement; the largest Holm-adjusted bootstrap value across all eight contrasts was 0.0020.

The waveform representations were not equal in width: after fixed per-file aggregation, qscore had 2 variables, the conventional domain bank 336, HCRD-1 534, and HCRD-8 2847. We therefore added a supplementary matched-capacity control, frozen before its scores were read, that expands the same 75 raw waveform inputs into Gaussian smoothing, first-derivative, and curvature channels and aggregates them to exactly 2847 variables. Its result is reported separately below because it was not part of the pre-specified E2 family.

Multilevel HCRD also exceeded one-level HCRD in both directions, by 0.0439 and 0.0544 AP; the second interval $[0.0076, 0.0955]$ remained positive after Holm correction ($p = 0.0372$), while the first $[-0.0167, 0.1060]$ did not. The best point representation was HCRD scalar geometry plus qscore (AP 0.9150) in the first direction and the multilevel area/energy spectrum plus qscore (0.9412) in the second. Conversely, full HCRD-8 did not establish a difference from the larger conventional domain bank: its two intervals were $[-0.0469, 0.0561]$ and $[-0.0479, 0.0154]$. Thus E2 confirms incremental multilevel convex-lobe information over the established minimal score, while the optimal compression of that hierarchy and superiority to every non-neural domain representation remain open.

The primary E2 bootstrap is conditional on the fitted source-study model. We therefore added two fixed secondary sensitivities. The first keeps the source model fixed and resamples target retention-time blocks, directly testing the conditional transfer estimand under local feature dependence. At the primary 60-second width, HCRD-8+Q minus qscore remained positive in both directions: 0.1181 ($[0.0130, 0.2190]$) and 0.1005 ($[0.0338, 0.1965]$). Both 30-second intervals were also positive; only the Falkor-to-MESOSCOPE interval crossed zero at the coarsest 120-second width. The second sensitivity resamples source and target blocks, refits the source learner in every replicate, and varies block width from 30 to 120 seconds. A separate ten-fold acquisition-file deletion recomputes the per-file representation aggregates before refitting both directions. These analyses propagate source-model, local retention-time, and acquisition-file sensitivity, but the public schema contains no compound/adduct identifiers and cannot support a chemical-cluster bootstrap.

Table 2: E2 dependence and refitting sensitivity for HCRD-8+Q minus qscore AP. Fixed-source rows report the original point difference and target RT-block percentile interval; source-refit rows give the bootstrap mean and interval. The file row gives the range across ten delete-group representation-and-model refits.

Design	Falkor → MESOSCOPE	MESOSCOPE → Falkor
Fixed-source target RT blocks (30 s)	0.1181 [0.0340, 0.2051]	0.1005 [0.0402, 0.1762]
Fixed-source target RT blocks (60 s)	0.1181 [0.0130, 0.2190]	0.1005 [0.0338, 0.1965]
Fixed-source target RT blocks (120 s)	0.1181 [-0.0227, 0.2418]	0.1005 [0.0415, 0.2082]
Source-refit RT blocks (30 s)	0.1087 [-0.0105, 0.2074]	0.0556 [-0.0671, 0.1602]
Source-refit RT blocks (60 s)	0.1000 [-0.0444, 0.2179]	0.0515 [-0.0935, 0.1742]
Source-refit RT blocks (120 s)	0.0886 [-0.1514, 0.2508]	0.0574 [-0.0803, 0.1761]
Acquisition-file delete-group	0.1030–0.1397 (10/10 positive)	0.0517–0.1058 (10/10 positive)

The source-refit mean remained positive in all six direction–block-width designs (positive paired replicates: 80.7–96.3%), but every percentile interval crossed zero. This secondary analysis therefore preserves the sign of the point effect without establishing it after source-model and RT-block resampling. The stronger file-deletion check was uniformly positive: the full-pipeline difference ranged from 0.1030 to 0.1397 in Falkor-to-MESOSCOPE and from 0.0517 to 0.1058 in the reverse direction over all 20 direction–fold fits.

In the matched-capacity sensitivity analysis, the Gaussian-derivative control had AP 0.9063 versus 0.8955 for HCRD-8 in Falkor-to-MESOSCOPE; the HCRD minus control difference was -0.0108 ($[-0.0594, 0.0405]$, Holm-adjusted $p = 0.7653$). In the reverse transfer, HCRD-8 had AP 0.8990 versus 0.7815, a difference of 0.1175 ($[0.0624, 0.1654]$, adjusted $p = 0.00040$). Thus raw dimensionality does not explain the reverse-direction gain, but the first direction provides no evidence that HCRD uniformly dominates a comparably rich conventional scale-space bank.

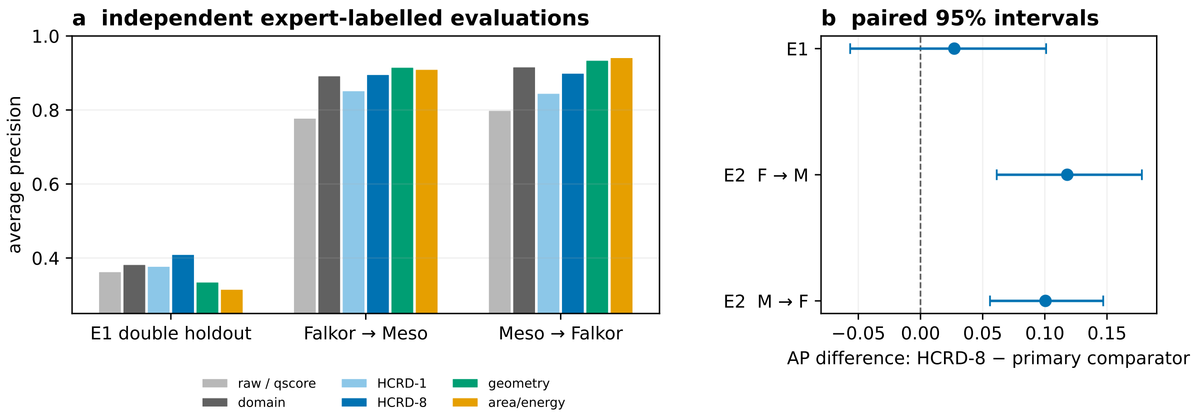

Figure 4: Independent expert-labelled LC–MS evidence. Panel a shows average precision for the first double holdout and both no-refit E2 transfers; "raw / qscore" denotes the task-appropriate minimal representation. Panel b shows the predeclared HCRD-8 difference from the primary comparator. The E1 interval resamples both sample and candidate-ion groups; E2 target-feature intervals are conditional on the trained source models. F and M abbreviate Falkor and MESOSCOPE.

LCMS-E5 tested a different operational consequence on Pttime. The source authors manually labelled only 400 of 7781 features: those already assigned a probability above 0.9 by their two-variable model. We therefore froze E5 as a conditional residual-error reranking test, not as a third fully labelled cohort. Before any target waveform was inspected, HCRD-8, the learner, AP-bad endpoint, and success rule were fixed. The Falkor and MESOSCOPE cases were pooled for training and the model was applied without refitting to the 365 unambiguous Pttime cases (348 Good, 17 Bad). Archive inspection before feature extraction revealed both ion modes; the source pipeline's POS filename schema fixed the compatible 52 positive-mode files, and this correction was recorded before any E5 score was computed.

Table 3 shows that HCRD-8 passed both pre-specified conditions. It attained AP-bad 0.5085 versus 0.0391 for qscore, a difference of 0.4694 with a paired class-stratified bootstrap interval $[0.2489, 0.6762]$ ($p = 0.00020$), and exceeded HCRD-1 by 0.2240 ($[0.0707, 0.3544]$, $p = 0.00060$). Seven of the 17 residual bad features were placed in the top 19 (5%) review positions, versus none under qscore. At 1%, 5%, and 10% queue budgets (4, 19, and 37 reviews), HCRD-8 retrieved 4, 7, and 8 bad cases, respectively, versus 0, 0, and 1 for qscore. Hence its 5%-budget precision was 36.8% and residual-bad recall was 41.2%. To reach at least 50% recall (9/17), HCRD-8 required 44 reviews versus 284 under qscore, an 84.5% workload reduction at the same integer recall target. HCRD-8 also had the best point estimate, but its 0.1212 AP advantage over DOMAIN+Q was not established ($[-0.0100, 0.2794]$, $p = 0.0818$). Thus E5 confirms practical conditional triage value and the need for multiple levels; selection of the labelled population prevents claims about all 7781 Pttime features, recall, calibration, or FDR.

Table 3: Frozen E5 Pttime residual-bad reranking within the 365 unambiguous source-model-selected features. AP-bad is primary; queue entries give bad features retrieved among 17 at review budgets of 4, 19, and 37 cases.

Representation	AP-bad	ROC AUC-bad	1%	5%	10%
qscore	0.0391	0.3403	0	0	1
DOMAIN+Q	0.3873	0.8115	4	6	6
HCRD-1+Q	0.2845	0.7792	2	5	7
HCRD-8+Q	**0.5085**	**0.8646**	**4**	**7**	**8**
HCRD geometry+Q	0.3540	0.8081	3	6	7
Area/energy+Q	0.1367	0.7194	1	3	5

E6 then tested a more severe external shift using the manually rated TARDIS FAME saliva panel [22]. The protocol, primary cohort, representations, pooled Falkor+MESOSCOPE learner, AP-bad endpoint, two co-primary comparisons, 20,000-replicate paired bootstrap, and seed were frozen before any rating row was inspected. The 5.67-GB Zenodo archive matched its published MD5. A pre-score availability audit found 119 positive-polarity QC mzXML files but no numerical negative-polarity EICs; the positive stratum was therefore evaluated and the negative stratum declared unavailable rather than approximated from diagnostic images. Fixed 10-ppm, 60-s boxes yielded valid feature banks for all 212 keyed positive targets, with median QC availability 0.9412; 108 Good and 84 Bad targets entered the binary endpoint and 20 Ambiguous targets only the ordered-label sensitivity analysis.

Table 4 reports the no-refit result. HCRD-8+Q improved AP-bad over qscore by 0.0725, but the percentile interval $[-0.0155, 0.1345]$ crossed zero ($p = 0.1115$), so the first co-primary rule failed. It was tied with HCRD-1+Q (difference -0.0016, $[-0.0500, 0.0453]$), so the second also failed. DOMAIN+Q had the largest AP point estimate, while HCRD geometry+Q had the largest ROC AUC. Thus decomposition-derived shape information transferred in point estimate beyond qscore, but E6 does not establish that eight levels are needed or that HCRD beats a compact conventional domain bank under this acquisition and processing shift.

Table 4: Frozen E6 no-refit transfer to 192 unambiguous positive-polarity TARDIS/FAME targets. AP-bad is primary; enrichment is the Bad fraction in the top score decile divided by overall Bad prevalence.

Representation	AP-bad	ROC AUC-bad	Top-decile enrich.
qscore	0.4353	0.5230	0.686
DOMAIN+Q	**0.5193**	0.6030	1.257
HCRD-1+Q	0.5094	0.5902	**1.486**
HCRD-8+Q	0.5078	0.5998	1.371
HCRD geometry+Q	0.5044	**0.6049**	**1.486**
Area/energy+Q	0.3847	0.4377	0.571

Two LC–MS boundary tests and five cross-domain searches delimit the claim. HCRD did not improve a morphology-saturated TargetedMSQC bank, and sign-aware lower-envelope estimators were better for chromatographic baseline recovery. It also did not establish superiority for specialist ECG/PPG delineation, bearing classification at ceiling, or real-KPI anomaly detection; full tables and the stability/persistence audits are in the supplement. These results support the narrower use of HCRD as a transparent representation for already bounded compact-lobe morphology. Figure 5 illustrates the complementary curvature- visibility failure behind one of these boundaries.

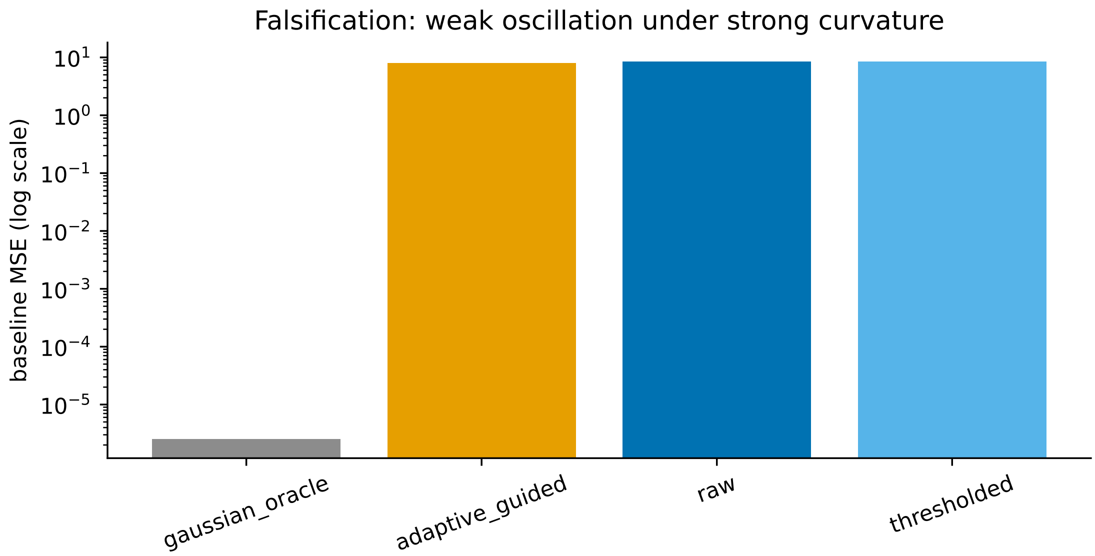

Figure 5: Limitation under strong background curvature. When background curvature exceeds the oscillatory curvature, raw HCRD cannot expose the oscillation and its chord baseline is inappropriate.

9 Discussion

HCRD occupies a specific rather than universal signal-processing niche. Its algorithmic appeal is its economy: divided-slope comparisons select maximal one-sign runs, endpoint chords form the next baseline, and subtraction gives an exactly reconstructing hierarchy. The resulting lobes expose support, sign, amplitude, residual shape, area, and level without a learned dictionary or a continuous smoothing parameter. This combination of simplicity and interpretability is the main contribution; its intended use is as a geometric representation of bounded waveform morphology, not as a replacement for every task-specific detector or decomposition.

The generative recovery result makes that niche testable. On alternating one-sign curvature blocks, the same geometry used by the algorithm yields exact first-level boundary recovery and explicit component-error control. The condition $\gamma > \eta + 2\tau_\delta$ allows nonzero sampled join curvature; the original noise-only threshold remains valid and does not require η. Across the exact- and approximate-join phase experiments, zero failures were observed among 80,000 draws in the strict certified regions. Outside the sufficient boundary, the less conservative noise-only threshold was often more accurate. This is a class-specific mechanism result, not a separation theorem against every adaptive representation.

The clearest practical evidence is expert curation of pre-bounded LC–MS features. In the two frozen no-refit E2 transfers, multilevel HCRD improved average precision over an independent reimplementation of the published two-component score by 0.1181 and 0.1005, with Holm-adjusted $p = 0.00040$ in both directions. HCRD-8 had positive point differences from the one-level variant in both transfers, with multiplicity-adjusted support in one. The equal-dimensional conventional control tied HCRD-8 in one transfer, whereas HCRD-8 was superior in the reverse. The incremental gain also survived four qscore implementations in both directions, so imperfect reproduction of one two-variable comparator is not an adequate explanation. It remained positive in all 20 direction–fold comparisons after acquisition-file deletion and full representation refitting. Fixed-source target RT-block intervals were positive in both directions at the primary 60-second width, so the conditional result is not an artefact of independent target-feature resampling. Source-refit RT-block means were also positive for all tested block widths, although their percentile intervals included zero. Hence the E2 claim remains conditional on the fitted source models rather than an unconditional population-level assertion. E5 further supports residual-error reranking when only conditional labels are available. E6, however, did not confirm superiority under the larger TARDIS/FAME laboratory shift: the gain over qscore was 0.0725 only in point estimate, HCRD-8 tied HCRD-1, and a compact conventional feature bank remained best. Thus the evidence supports transferable value for pre-bounded compact-lobe curation, with hierarchy depth still dependent on the acquisition domain.

Results outside pre-bounded compact-lobe curation sharpen that positioning. HCRD added no benefit after a morphology-saturated LC–MS feature bank (E3), and sign-specific lower-envelope estimators were better chromatographic baseline models (E4). The ECG-specific `ecgpuwave` system retained lower QTDB boundary error; tuned `find_peaks` remained better overall on wearable PPG; and the supervised UCR, real WSD/AIOps anomaly, and bearing-RUL studies did not establish an advantage. These are important boundaries rather than a contradiction: a small generic geometry should not be expected to dominate specialist multichannel morphology rules, correct sign priors, or large supervised representations on their native objectives.

The empirical and theoretical results together suggest a practical selection rule. HCRD is most plausible when the object is already bounded, an endpoint chord is a meaningful local reference, and scientifically relevant structure appears as signed convex or concave excursions whose complete hierarchy may matter. It is less well matched to frequency separation, instantaneous frequency estimation, weak oscillations on a strongly curved trend, or noisy samplewise decisions without thresholding. Polygon mass and its levelwise time series are inexpensive auxiliary summaries, but the PPG and RUL evidence does not support replacing the full decomposition by area alone.

The simplicity also has computational consequences. With sparse retained knots and the halving property, total hierarchy work is linear in signal length; dense materialization can remain $O(n \log n)$. On P2R the knot-only implementation was 11.89-fold faster than dense materialization while preserving knots and downstream features bitwise. Independent signals admit exact process parallelism, including a 1.96-fold median speedup on the 3840-signal sparse P3 batch. Absolute speedups remain hardware- and batch-dependent, and the present Python implementation does not establish an optimal systems realization.

The theory states the same boundary precisely. A hard knot map cannot be globally continuous while preserving every generic raw branch decision (Theorem 6); persistence matching and the proximal outer split provide well-posed stable alternatives, while deterministic and stochastic margins certify local hard-HCRD decisions. The explicit chord-lobe theorem turns those decision margins into boundary and component recovery on a declared generative class. Optional rank-aware scans and independent-guide inference in the supplement provide calibrated uses of the resulting shapes. These results justify a signed curvature-lobe representation, but do not identify its details with intrinsic modes.

The remaining gaps are consequently focused. The theory still assumes known or conservatively supplied noise scale and, for optional inference results, covariance, dependence lag, and stability radius. Estimated versions, infinite-memory noise, unsampled transitions with quantitative localisation, and general polygonal templates remain open. A minimax separation from all competing adaptive dictionaries is not claimed. Empirically, representation selection must be frozen outside E2, and a further waveform-backed cross-laboratory LC–MS cohort is needed to resolve the roughly 0.07 AP effect seen in E6 and determine when multilevel depth transfers. The present evidence therefore supports a task-specific method with unusually transparent mechanics and failure modes. Broader state-of-the-art performance requires future independent confirmation.

Code and data availability

Implementation, tests, protocols, compact outputs, hashes, and environment instructions are at `https://github.com/Safreliy/hcrd`. Large derived caches are rebuilt locally.

CRediT authorship contribution statement

Saveliy Baturin: Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Visualization; Project administration. Writing–original draft; Writing–review and editing.

Funding

This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have influenced the work reported in this article.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI Codex to assist with literature organization, language and clarity editing, and manuscript formatting. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

[1] Roumen Anguelov and Inger Fabris-Rotelli. Lulu operators and discrete pulse transform for multidimensional arrays. *IEEE Transactions on Image Processing*, 19(11):3012–3023, 2010. doi: 10.1109/TIP.2010.2050639.

[2] Jean Babaud, Andrew P. Witkin, Michel Baudin, and Richard O. Duda. Uniqueness of the gaussian kernel for scale-space filtering. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-8(1):26–33, 1986. doi: 10.1109/TPAMI.1986.4767749.

[3] Christoph Bueschl, Maria Doppler, Elisabeth Varga, Bernhard Seidl, Mira Flasch, Benedikt Warth, and Juergen Zanghellini. PeakBot: Machine-learning-based chromatographic peak picking. *Bioinformatics*, 38 (13):3422–3428, 2022. doi: 10.1093/bioinformatics/btac344.

[4] Anatole Chessel, Bertrand Cinquin, Sabine Bardin, Jean Salamero, and Charles Kervrann. Computational geometry-based scale-space and modal image decomposition. In *Scale Space and Variational Methods in Computer Vision*, pages 770–781, 2009. doi: 10.1007/978-3-642-02256-2_64.

[5] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007. doi: 10.1007/s00454-006-1276-5.

[6] Éric Deléchelle, Jacques Lemoine, and Oumar Niang. Empirical mode decomposition: An analytical approach for sifting process. *IEEE Signal Processing Letters*, 12(11):764–767, 2005. doi: 10.1109/LSP.2005.856878.

[7] David H. Douglas and Thomas K. Peucker. Algorithms for the reduction of the number of points required to represent a digitized line or its caricature. *Cartographica: The International Journal for Geographic Information and Geovisualization*, 10(2):112–122, 1973. doi: 10.3138/FM57-6770-U75U-7727.

[8] Mark G. Frei and Ivan Osorio. Method, computer program, and system for intrinsic timescale decomposition, filtering, and automated analysis of signals of arbitrary origin or timescale. US Patent 7,054,792, 2006. URL `https://patents.google.com/patent/US7054792B2/en`. Priority 11 October 2002; issued 30 May 2006.

[9] Mark G. Frei and Ivan Osorio. Intrinsic time-scale decomposition: time–frequency–energy analysis and real-time filtering of non-stationary signals. *Proceedings of the Royal Society A*, 463:321–342, 2007. doi: 10.1098/rspa.2006.1761.

[10] Norden E. Huang, Zheng Shen, Steven R. Long, Manli C. Wu, Hsing H. Shih, Quanan Zheng, Nai-Chyuan Yen, Chi Chao Tung, and Henry H. Liu. The empirical mode decomposition and the hilbert spectrum for nonlinear and non-stationary time series analysis. *Proceedings of the Royal Society A*, 454:903–995, 1998. doi: 10.1098/rspa.1998.0193.

[11] Byeong-Gwan Iem. A nonuniform sampling technique based on inflection point detection and its application to speech coding. *Journal of the Acoustical Society of America*, 136(2):903–909, 2014. doi: 10.1121/1.4884882.

[12] Mohsen Kafil, Kaveh Darabi, and Saeed Ziaei-Rad. An improved EMD method based on utilizing certain inflection points in the construction of envelope curves. *Archives of Acoustics*, 48(3):389–401, 2023. doi: 10.24425/aoa.2023.145245.

[13] Seung-Jean Kim, Kwangmoo Koh, Stephen Boyd, and Dimitry Gorinevsky. ℓ_1 trend filtering. *SIAM Review*, 51(2):339–360, 2009. doi: 10.1137/070690274.

[14] William Kumler, Bryna J. Hazelton, and Anitra E. Ingalls. Picky with peakpicking: Assessing chromatographic peak quality with simple metrics in metabolomics. *BMC Bioinformatics*, 24(1):404, 2023. doi: 10.1186/s12859-023-05533-4.

[15] Dirk P. Laurie. The roadmaker's algorithm for the discrete pulse transform. *IEEE Transactions on Image Processing*, 20(2):361–371, 2011. doi: 10.1109/TIP.2010.2057255.

[16] Luan Lin, Yang Wang, and Haomin Zhou. Iterative filtering as an alternative algorithm for empirical mode decomposition. *Advances in Adaptive Data Analysis*, 1(4):543–560, 2009. doi: 10.1142/S1793536909000280.

[17] Erik Müller, Carolin Huber, Liza-Marie Beckers, Werner Brack, Martin Krauss, and Tobias Schulze. A data set of 255,000 randomly selected and manually classified extracted ion chromatograms for evaluation of peak detection methods. *Metabolites*, 10(4):162, 2020. doi: 10.3390/metabo10040162.

[18] Neal Parikh and Stephen Boyd. Proximal algorithms. *Foundations and Trends in Optimization*, 1(3):127–239, 2014. doi: 10.1561/2400000003.

[19] Philippe Salembier, Albert Oliveras, and Luis Garrido. Antiextensive connected operators for image and sequence processing. *IEEE Transactions on Image Processing*, 7(4):555–570, 1998. doi: 10.1109/83.663500.

[20] Ethan Stancliffe and Gary J. Patti. PeakDetective: A semisupervised deep learning-based approach for peak curation in untargeted metabolomics. *Analytical Chemistry*, 95(25):9397–9403, 2023. doi: 10.1021/acs.analchem.3c00764.

[21] Cole van Jaarsveldt, Matthew Ames, Gareth W. Peters, and Mike Chantler. Package AdvEMDpy: Algorithmic variations of empirical mode decomposition in python. *Annals of Actuarial Science*, 17(3):606–642, 2023. doi: 10.1017/S1748499523000088.

[22] Pablo Vangeenderhuysen, Matthijs Vynck, Beata Pomian, Kimberly De Windt, Emile Callemeyn, Ellen De Paepe, Lindsey De Commer, Jeroen Raes, Tim Nawrot, Johannes Rainer, Lieselot Y. Hemeryck, and Lynn Vanhaecke. Automated integration and quality assessment of chromatographic peaks in LC–MS-based metabolomics and lipidomics using TARDIS. *Analytical Chemistry*, 97(18):9927–9934, 2025. doi: 10.1021/acs.analchem.5c00567.

[23] Chi Yang, Yung-Chin Hsiao, Chi-Ching Lee, and Jau-Song Yu. Systematic evaluation of chromatographic peak quality for targeted mass spectrometry via variational autoencoder. *Analytical Chemistry*, 96(7):2849–2856, 2024. doi: 10.1021/acs.analchem.3c03686.

[24] Huaxu Yu, Jun Ding, Tong Shen, Min Liu, Yuanyue Li, and Oliver Fiehn. MassCube improves accuracy for metabolomics data processing from raw files to phenotype classifiers. *Nature Communications*, 16:5487, 2025. doi: 10.1038/s41467-025-60640-5.